



Personal Data

Family Name:
Given Name:
Signature:
checked

Registration Number

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0	☐	☐	☐	☐	☐	☐	☐	0
1	☐	☐	☐	☐	☐	☐	☐	1
2	☐	☐	☐	☐	☐	☐	☐	2
3	☐	☐	☐	☐	☐	☐	☐	3
4	☐	☐	☐	☐	☐	☐	☐	4
5	☐	☐	☐	☐	☐	☐	☐	5
6	☐	☐	☐	☐	☐	☐	☐	6
7	☐	☐	☐	☐	☐	☐	☐	7
8	☐	☐	☐	☐	☐	☐	☐	8
9	☐	☐	☐	☐	☐	☐	☐	9

In this section **no** changes or modifications must be made!

Scrambling

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Type

001

Exam ID

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Only clearly marked and positionally accurate crosses will be processed!

Answers 1 - 1

	a	b	c	d	e
1	☐	☐	☐	☐	☐
	a	b	c	d	e



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1	☐	☐	☐	☐	☐	☐	☐	1
2	☐	☐	☐	☐	☐	☐	☐	2
3	☐	☐	☐	☐	☐	☐	☐	3
4	☐	☐	☐	☐	☐	☐	☐	4
5	☐	☐	☐	☐	☐	☐	☐	5
6	☐	☐	☐	☐	☐	☐	☐	6
7	☐	☐	☐	☐	☐	☐	☐	7
8	☐	☐	☐	☐	☐	☐	☐	8
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Answers 1 - 1

	a	b	c	d	e
1	☐	☐	☐	☐	☐
	a	b	c	d	e



1. According to the Lewis definition of acid-base reactivity, which of the following statements correctly defines a Lewis acid?
- a. A chemical species that increases the concentration of hydroxide ions (OH^-) when dissolved in an aqueous solvent medium.
 - b. Any chemical species bearing a net positive formal charge, such as the tetramethylammonium cation ($(\text{CH}_3)_4\text{N}^+$), regardless of valence octet occupancy.
 - c. A chemical species that donates an unshared electron pair from a filled nonbonding orbital to form a coordinate covalent bond.
 - d. A chemical species that accepts an electron pair to form a new coordinate covalent bond using a vacant low-energy orbital.
 - e. A chemical species that donates a proton (H^+) via heterolytic cleavage of a polarized $\text{X} - \text{H}$ bond.